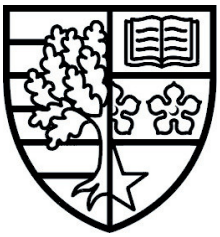
Skin Lesion Classification on FPGA

by Lewis Turner

Research Report

Submitted in partial fulfilment for the degree of
BSc (Hons) Computer Science

Supervised by
Dr. Rob Stewart
Ms. Tess Watt



Heriot-Watt University
School of Mathematical and Computer Sciences

7th November 2026

The copyright in this dissertation is owned by the author. Any quotation from the dissertation or use of any of the information contained in it must acknowledge it as the source of the quotation or information.

Declaration of Authorship

Course code and name	F20PA/B/C - Honours Project
Type of assessment	Individual
Document title	Skin Lesion Classification on FPGA
Student name	Lewis Turner
Student ID	H00430662

Statement on Generative AI Use:

• Google Gemini 3.8 Flash (Extended Thinking) to assist in searching for relevant research papers.

• Google Gemini 4.0 Argon (thinking: Low - High) to assist setting up FPGA for use over Wifi.

By signing this form:

• I declare that the work I have submitted for this assessment is my own original work. I have not presented the ideas, words, data, code, designs, analyses, or other intellectual contributions of another person or system as my own.

• My submission is expressed in my own words except where the work of others has been appropriately acknowledged, cited, and referenced. All sources used in the preparation of this assessment, including books, journal articles, websites, software tools, and Generative AI tools, have been properly acknowledged.

• Where Generative AI tools have been used, I have complied with the course's stated policies regarding Generative AI use. I have clearly and accurately disclosed in the above statement box all uses of Generative AI in the assessment, including the specific tools used, the purpose for which they were used, and the extent to which any generated content influenced the final submission.

• I understand that I remain fully responsible for the content, accuracy, integrity, and originality of my submission, regardless of any assistance received from Generative AI or other tools.

• I confirm that I have read, understood and followed the University's Regulations on academic misconduct as published on the University's website, and that I am aware of the penalties that I will face should I not adhere to the University Regulations.

(form continues on next page)

(form continued from previous page)

• I confirm that I have read, understood and avoided the different types of plagiarism explained in the University guidance on Academic Integrity and Plagiarism.

Student signature: Lewis Turner

Date: 7th November 2026

Acknowledgements

Acknowledgements text.

Abstract

Abstract text.

Table of Contents

Declaration of Authorship i
Acknowledgements iii
Abstract iv
Table of Contents v
1 Introduction 1
1.1 Background and Motivation . . . 1
1.2 Aim and Objectives . . . 1
2 Literature Review 2
2.1 Acknowledgements . . . 2
2.2 Search Methods . . . 2
2.3 Machine Learning Models on FPGA . . . 2
2.4 Compressing Machine Learning Models to FPGA . . . 2
2.5 Skin Lesion Classification Models . . . 2
2.6 Further Research Recommendations . . . 3
3 Methodology 3
3.1 Requirements . . . 3
3.2 Plan for Execution . . . 3
3.3 Experiment Planning . . . 3
4 Feasibility 4
4.1 Preliminary Work . . . 4
4.2 Evaluation Strategy . . . 4
5 Conclusions 4
References 5
6 Project Management 5
6.1 Gantt Chart . . . 5
6.2 Analysis of Risks . . . 5
6.3 PLES . . . 5
7 Gen AI Summary 5

1 Introduction

Summary of project motivation & context

1.1 Background and Motivation

- Using papers (mainly) from literature review to provide background context
- Motivations for doing this project
- Identifying problems to be solved to achieve objectives

1.2 Aim and Objectives

State project aims
Table of objectives with IDs to reference later

2 Literature Review

The proposal for this project was to run a skin lesion classification model on an FPGA. I chose the Xilinx PYNQ-Z1 FPGA board as my target hardware because any FPGA board would have been appropriate for this project and my supervisor Tess Watt had one available by donation to Heriot-Watt University from AMD.
The FPGA board has significant memory and resource constraints [2] so Section 2.4 is to find suitable ML compression methods for FPGAs that best retain baseline model performance and best utilise the FPGA.
ML models are deployed to the FPGA using a pipeline framework, Section 2.3 looks into multiple methods for deploying a model to FPGA with different benefits and supported model types.
The final area of research in Section 2.5 is finding existing skin lesion classification models, both unrelated to FPGAs and designed around running on an FPGA.

2.1 Acknowledgements

Sources for literature/statements
Continuing idea by supervisor Tess Watt in her paper but deployed to FPGA instead of RPi [5]
Survey of recent methods for skin lesion classification between 2017 and 2023 [1]

2.2 Search Methods

I used Google Gemini 3.8 Flash on thinking mode High (October 2026 version) to source the most recent academic papers on a particular topic. I would ask it to create a table consisting of the papers: title, year, summary, DOI or similar URL. This allowed me to leverage the LLM's speed for searching the internet like a more powerful keyword search.
I also used Google Scholar to search for papers by particular keywords such as "skin lesion classification".

2.3 Machine Learning Models on FPGA

Skin lesion classification models are image classifier models. This section describes a few options for pipelines that could be used.

Explain here details about FINN framework: general process, benefits, drawbacks, supported model types

2.4 Compressing Machine Learning Models to FPGA

Papers on useful compression techniques for certain model types

2.5 Skin Lesion Classification Models

using ML has greatly improved early detection of skin lesions...

What is a skin lesion classification model?
Type of Machine Learning model trained to identify lesion areas in images of skin areas

[4, 5].
Choosing an appropriate pre-existing model or training my own will determine the baseline performance I will evaluate against later on.

Top datasets:
Ham10000 - contains 10,015 dermoscopic (medical, of skin area) images categorised for 7 different classes of skin lesion. Used as a standardised benchmark for multi-class skin lesion classification because data is verified to very high standard [1].
ISIC Archive and Challenge - multiple releases of thousands of curated dermoscopic images. Used as a standardised competition benchmarks with split training/validation/test sets and evaluation protocols [1].
PH2 - small set containing 200 dermoscopic images with 3 different classes. Used because includes expert manual verification.

Existing models: Multi-modal approach on an FPGA which combines image features with patient metadata [4]. Uses MobileNetV2 architecture.

Architectures:
Fundamental MobileNet paper with X features [3].

2.6 Further Research Recommendations

Identify gaps, need to repeat any research

3 Methodology

Detailed description of necessary methods, requirements, research questions that will be employed/ needed to complete the project

3.1 Requirements

MOSCOW
Enumerated to be easier to reference again

3.2 Plan for Execution

Explain how I plan to use experiments to fulfil all the requirements & objectives

3.3 Experiment Planning

How I plan to conduct experiment(s) and what research question does it address/answer (detailed)

4 Feasibility

4.1 Preliminary Work

Summary of work carried out to prepare for this project to validate feasibility.

4.2 Evaluation Strategy

Identify for the project:
• What needs evaluating
• Why it should be evaluated
• How it should be evaluated

5 Conclusions

Short summary of this doc

References

[1] Taye Girma Debelee. 2023. Skin Lesion Classification and Detection Using Machine Learning Techniques: A Systematic Review. *Diagnostics* 13, 19 (2023). doi:10.3390/diagnostics13193147

[2] Diligent, Inc. [n. d.]. *PYNQ-Z1 Reference Manual.* https://digilent.com/reference/programmable-logic/pynq-z1/reference-manual Accessed: 2026-10-20.

[3] Andrew G. Howard, Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. 2017. MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. arXiv:1704.04861 [cs.CV] https://arxiv.org/abs/1704.04861

[4] Zhihao Mao, Jin Han, and Mingzhu Shi. 2025. Design and Implementation of a Real-Time Embedded System for Early Skin Cancer Detection. *Journal of Engineering and Science in Medical Diagnostics and Therapy* 9, 2 (12 2025), 021202. arXiv:https://asmedigitalcollection.asme.org/medicaldiagnostics/article-pdf/9/2/021202/7529299/jesmdt-25-1029.pdf doi:10.1115/1.4069441

[5] Tess Watt, Christos Chrysoulas, Peter J. Barclay, Brahim El Boudani, and Grigorios Kalliatakis. 2024. Moving Healthcare AI Support Systems for Visually Detectable Diseases to Constrained Devices. *Applied Sciences* 14, 24 (2024). doi:10.3390/app142411474

6 Project Management

6.1 Gantt Chart

Plan of activities to be carried out in semester 1, then future plan for semester 2

6.2 Analysis of Risks

Including mitigation plans

6.3 PLES

A well-researched consideration of any Professional, Legal, Ethical, and Social Issues pertinent to the project. (e.g. codes of conduct (BCS), codes of practice, standards, computer law, ethical decision making, intellectual property, social aspects, copyright, data protection, and so on).

7 Gen AI Summary

Summarising use of GenAI in project/deliverables